

Fruit Fly

Team: Cheesecake

Element B: Documentation and Analysis of Prior Design Attempts

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

A window fly trap is placed on a window and aims with its sticky surface to capture flies that are trying to fly out (Amazon). Positive aspects of this previous design are that it has an easy installation and has a relatively low price of \$10.33 for 12 traps.



Negative aspects of the window fly trap are that it is a mess if touched and needs to be maintained often because of the fact that each of the flies have to be physically removed from the trap, or the trap has to be replaced.

Essential Oils in a container placed in a room can help to remove flies. The fruit flies do not like the strong fragrances from the herbs or oils. Positive aspects of essential oils are that it has a natural odor dampener (Maid2Match). Also, it is easy to maintain because there are no flies that are to be removed. Negative aspects are that essential oils have an odor which could bother certain



people.

An apple shaped trap that has a substance attracting flies has also been a previous solution. This specific product claims that it prevents flies from reproducing and multiplying, and its smell is not strong, just enough to remove flies (USA Today). It is also easy to activate the traps. Some negative aspects of this solution is that it has a low capacity because the “apples” are small and will only hold a low amount of trapped flies. This product is also sensitive to temperature changes because it is plastic which amplifies the chance of it breaking if touched.



Our stakeholder is currently not incorporating any solution to the problem. This approach also has a small positive aspect being that no monetary purchases or time and labor were applied as well as there being no maintenance as preferred in the design requirements.

Works Cited

Amazon. Com: Window fly traps by catchmaster—12 count, ready to use indoors. Insect, bugs, fly & fruit fly glue adhesive sticky paper—Waterproof easy application ready disposable non-toxic: Home pest control traps: Patio, lawn & garden. (n.d.). Retrieved December 1, 2023, from https://www.amazon.com/CatchMaster-904-12-Clear-Window-24-Pack/dp/B001QBPP66/ref=as_li_ss_tl?ots=1&slotNum=3&imprToken=32c610dd-89b4-889b-003&ie=UTF8&linkCode=ll1&tag=rwwfeature-20&linkId=c1e6cecba6b81dd59d05218a1fe800ad&language=en_US&th=1

Brezinski, M., Testing, S. W. W. B., August 27, reviewing products for over a decade U., &

2019. (n.d.). *8 products that actually get rid of fruit flies, according to reviewers.*

Reviewed. Retrieved December 1, 2023, from

<https://reviewed.usatoday.com/home-outdoors/features/8-products-that-actually-get-rid-of-fruit-flies>

Franklin, S. (2022, July 31). 7 friendly ways to get rid of houseflies. *Maid2Match*.

<https://www.maid2match.com.au/7-friendly-ways-to-get-rid-of-houseflies/>

How to get rid of fruit flies. (n.d.). The Home Depot. Retrieved December 1, 2023, from

<https://www.homedepot.com/c/ab/how-to-get-rid-of-fruit-flies/9ba683603be9fa5395fab90e5719e17>